

S.A. ENGINEERING COLLEGE

Autonomous Institution | Affiliated to Anna University | NBA & NAAC 'A' Grade

Dept of Electronics & Communication Engg & Dept of Electronics Engg [VLSI Design & Technology]

KERNEL PRIME '26 - SOFTWARE TRACK

National Level 24-Hour Hackathon | October 8-9, 2026

OFFICIAL PROBLEM STATEMENT: PS-06

Satellite & IoT Precision Agriculture Hydro-Optimization System

Domain: Agritech / Remote Sensing / Embedded Systems

1. PROBLEM OVERVIEW & CHALLENGE STATEMENT

Engineer an intelligent multi-spectral satellite imagery and soil telemetry pipeline that optimizes variable-rate irrigation scheduling and detects crop stress patterns 10 days before visible emergence.

2. MANDATORY ARCHITECTURAL REQUIREMENTS

[REQ-1] Sentinel-2 / Landsat NDVI and NDWI multi-spectral index calculation pipeline.

[REQ-2] LoRaWAN low-power mesh sensor network for soil volumetric water content and temperature.

[REQ-3] Evapotranspiration (Penman-Monteith) estimation combining local microclimate weather models.

[REQ-4] Automated solenoid valve automation control with solar battery power management.

[REQ-5] Farmer-first multilingual SMS/WhatsApp alert interface and offline-first PWA dashboard.

3. HACKATHON EVALUATION MATRIX

* System Architecture & Concurrency Resilience: 25%

* Technical Innovation & Production Viability: 25%

* Live Demo & Real-World Latency Benchmarks: 30%

* Security Hardening, UI Polish & Documentation: 20%

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